

Data Analytics Capstone

**Data-Driven Analysis and Predictive Modeling of Graduate
Employment and Salary Outcomes in Singapore**

Final Report

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QM640: Data Analytics Capstone

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GitHub Repository and Reproducibility Artifacts

GitHub repository for evaluator access:

https://github.com/sundaramkumaran/QM640_GES_Capstone

The repository provides the raw dataset, processed datasets, analysis notebook, output tables, figures, and report artifacts needed to reproduce the final analysis. Key local project paths are listed below and should be updated only if the project is moved to another computer or repository folder.

Table 1. Project artifact locations

Artifact	Path
Raw data	data/raw/graduate_employment_survey.csv
Processed data	data/processed/ges_cleaned.csv
Clustered processed data	data/processed/ges_clustered_program_profiles.csv
Notebook	notebooks/QM640_GES_Final_Master_Analysis.ipynb
Output tables	outputs/tables/
Output figures	outputs/figures/
Report package	outputs/report_package/
Full checkpoint	checkpoints/checkpoint_full_analysis_completed.pkl

Abstract

Graduate employment outcomes are often reviewed using isolated indicators such as median salary or overall employment rate. This creates an interpretation gap because program outcomes are multidimensional and may differ across salary level, stable employment, salary spread, university context, and year-to-year change. This study develops a reproducible analytics framework to explain, classify, and segment Singapore university graduate outcomes using official program-level Graduate Employment Survey data. The data include 1,550 university-degree-year records from 2013 to 2024 across NTU, NUS, SIT, SMU, SUSS, and SUTD, with 1,445 complete-case records used for the main statistical and modeling analysis. The solution combines descriptive trend analysis, Kruskal-Wallis testing, post-hoc pairwise comparison, linear regression, leakage-controlled classification, K-Means clustering, and a validated Career Value Index. The implementation uses Python, Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Jupyter Notebook, and GitHub-based reproducibility artifacts. No deep learning or agentic model is used because the task is a structured tabular analytics problem and interpretability is more important than model complexity. The analysis shows that median gross monthly salary increased from 3,100 in 2013 to 4,400 in 2024, while full-time permanent employment dropped sharply to 70.50% in 2020 before recovering. University-level differences were statistically significant for overall employment, full-time permanent employment, median salary, and Career

Value Index, although effect sizes were modest. The salary regression model achieved MAE = 333.86, RMSE = 476.90, and R-squared = 0.58. Leakage-controlled high-salary classification was the strongest predictive result, with F1 = 0.71 and ROC AUC = 0.91 under the stricter degree-grouped context-only split. K-Means clustering selected three interpretable employability-salary profiles with silhouette score 0.32. The Career Value Index was statistically supported by significant correlations and remained robust under alternative weighting schemes, with Spearman correlations from 0.98 to 1.00. The framework can support students comparing programs, universities monitoring graduate outcomes, and policymakers tracking labor-market alignment. The final contribution is a practical, reproducible decision-support framework that combines statistical evidence, model validation, segmentation, and CVI robustness testing in one workflow.

Introduction

Background and Context

Graduate employment outcomes are important public indicators of how effectively higher education programs connect students with labor-market opportunities. In Singapore, the Graduate Employment Survey (GES) provides program-level information on employment rates and salary outcomes across autonomous universities. The dataset is relevant for students comparing degree pathways, universities reviewing program outcomes, and policymakers monitoring labor-market alignment.

This project focuses on official program-level GES data for Nanyang Technological University (NTU), National University of Singapore (NUS), Singapore Institute of Technology (SIT), Singapore Management University (SMU), Singapore University of Social Sciences (SUSS), and Singapore University of Technology and Design (SUTD). The current analysis uses program-year records from 2013 through 2024.

Graduate outcomes should not be interpreted using salary alone. A program can have high median salary but wider salary dispersion, lower full-time permanent employment, or greater year-to-year variation. Similarly, a program can show high overall employment but lower stable employment. These differences are important because students and institutions need to distinguish general employment from stable employment and salary level from salary consistency.

This final report presents a complete analytics workflow from data preparation through exploratory analysis, statistical testing, regression, leakage-controlled classification, clustering, and Career Value Index validation. It explains the project, methods, results, contribution, limitations, and decision-support value.

Problem Statement

The objective of this study is to explain, classify, and segment graduate employment and salary outcomes for Singapore university program-year records using university, school, degree, year, employment, salary, salary-spread, and engineered career-value features from 2013 to 2024. Success is evaluated using statistical significance tests, regression metrics including MAE, RMSE, and R-squared, classification metrics including F1-score and ROC AUC, clustering metrics including silhouette score, and robustness validation of the Career Value Index.

Purpose of the Study

The purpose of this study is to analyze Singapore university program-level graduate outcomes using descriptive, inferential, predictive, and segmentation methods. The study aims to identify employment and salary trends, test whether outcomes differ across universities, examine factors associated with salary and Career Value Index, classify high-outcome program-year records, and segment programs into meaningful employability-salary profiles.

Research Questions

1. RQ1: How have employment and salary outcomes evolved across Singapore university degree programs from 2013 to 2024?
2. RQ2: Do employment and salary outcomes differ significantly across universities and degree programs?
3. RQ3: Which factors contribute most to graduate salary outcomes and graduate career value across Singapore university programs?
4. RQ4: Can machine learning models accurately predict high-employment, high-salary, and high-career-value degree programs?
5. RQ5: Can degree programs be segmented into meaningful employability-salary profiles to support student, university, and policy decision-making?

Contribution to Multidimensional Graduate Outcome Analysis

This project contributes a complete program-level analytics framework for interpreting Singapore graduate employment and salary outcomes. The practical contribution is a decision-support approach that helps students, universities, and policymakers compare degree program outcomes beyond salary alone. The technical contribution is the integration of trend analysis, statistical testing, regression modeling, leakage-controlled classification, K-Means clustering, and a validated Career Value Index.

The Career Value Index summarizes multiple graduate outcome dimensions, including overall employment, full-time permanent employment, median salary, and salary stability. The index is statistically validated through correlation analysis, diagnostic modeling, and sensitivity analysis. The final output is a reproducible Python-based workflow supported by GitHub artifacts, report-ready tables, figures, and appendices.

Literature Review

Literature Review Approach

The literature review focused on graduate employability, statistical learning, regression, classification, clustering, multivariate analysis, model evaluation, and sample-size adequacy. Sources were selected because they support the conceptual framing of employability, the statistical basis for the analysis, or the modeling techniques used in the notebook. The official MOE dataset source is included because it is the data foundation for all research questions.

Table 2. Literature relevance matrix

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it supports RQ(s) / project decisions
MOE (2026)	Singapore graduate outcomes	Official Singapore GES program-level dataset	Public administrative dataset	Provides employment and salary outcomes by university, school, degree, and year.	Primary data source for all RQs. Supports trend, university comparison, modelling, clustering, and CVI construction.
Tomlinson (2012)	Graduate employability	Higher education employability literature	Conceptual review	Employability is multidimensional and should not be reduced to one outcome measure.	Supports the project decision to evaluate salary, employment, stable employment, and CVI together.
Fugate et al. (2004)	Employability construct	Employability theory and career context	Theoretical model	Employability includes multiple personal and contextual dimensions.	Supports interpreting graduate outcomes beyond salary-only indicators.
Yorke (2006)	Employability in higher education	Higher education employability setting	Conceptual framework	Employability is broader than immediate employment or salary outcomes.	Supports the multidimensional framing used in RQ1, RQ3, and CVI design.
Shmueli (2010)	Analytics purpose	Statistical modelling and analytics research	Explain-versus-predict framework	Explanation and prediction have different goals and should be evaluated differently.	Supports separating salary regression as explanatory and classification as predictive.
James et al. (2023)	Statistical learning	Applied statistical learning examples	Regression and classification methods	Regression and classification require appropriate validation and performance metrics.	Supports model selection, train-test split, MAE, RMSE, R-squared, F1, and ROC AUC.
Breiman (2001)	Machine learning	Supervised learning / ensemble modelling	Random Forest	Ensemble methods can improve predictive performance but require careful validation.	Supports including Random Forest as a classification comparison model.
Pedregosa et al. (2011)	Machine learning software	Python scikit-learn ecosystem	Software library / model implementation	scikit-learn provides reproducible tools for preprocessing, modelling, and evaluation.	Supports the technical implementation of regression, classification, clustering, and pipelines.
MacQueen (1967)	Clustering	Numerical data segmentation	K-Means clustering	K-Means groups records into clusters based on feature similarity.	Supports RQ5 segmentation of employability-salary profiles.

Author (Year)	Domain / Context	Dataset / Setting	Method(s)	Key Findings	How it supports RQ(s) / project decisions
Jolliffe & Cadima (2016)	Multivariate analysis	PCA and dimensionality reduction research	PCA review	PCA helps summarize high-dimensional numeric structure for visualization.	Supports PCA visualization of K-Means clusters.
Green (1991)	Regression design	Regression sample-size guidance	Sample-size rule of thumb	Regression analysis requires adequate sample size relative to predictors.	Supports sample-size adequacy discussion for RQ3.
Hair et al. (2019)	Multivariate statistics	Applied multivariate analysis	Regression, correlation, interpretation	Multivariate models require careful interpretation of relationships and diagnostics.	Supports regression interpretation, correlation validation, and non-causal explanation.

The literature supports the project in three main ways. First, employability research emphasizes that graduate outcomes are not limited to salary. Employment stability and career readiness must also be considered. This aligns with the project decision to include full-time permanent employment, salary spread, and Career Value Index.

Second, predictive analytics literature distinguishes between models used for explanation and models used for prediction. This distinction is important because the salary regression model is interpreted as an association model, while the classification models are evaluated using prediction metrics. The Career Value Index regression is treated as diagnostic because the index is constructed from its component variables.

Third, clustering and multivariate analysis literature supports the segmentation phase. K-Means clustering is appropriate for creating groups from standardized numeric outcome dimensions. PCA is used only for visualization, while cluster interpretation is based on the original employment and salary variables.

Deep learning and agentic or large-language-model techniques were not selected for the final modeling pipeline because the available data is structured, tabular, and aggregated at the program-year level. The project prioritizes interpretability, statistical transparency, and reproducibility over black-box model complexity.

Materials and Methods

Project Workflow and Original Contribution

The original work in this project is a reproducible analytics workflow for studying Singapore graduate employment and salary outcomes at the program-year level. The workflow starts from the official Graduate Employment Survey dataset and produces cleaned data, engineered

features, statistical results, predictive models, clustering profiles, CVI validation outputs, and report-ready tables and figures.

The project uses a file-based workflow instead of a separate database server. The raw dataset, cleaned dataset, clustered dataset, notebook, figures, and output tables are organized in the GitHub repository so that the analysis can be reviewed and reproduced.

The data preparation work included standardizing column names, converting employment and salary fields to numeric format, handling placeholders such as “na” and “N.A.”, checking duplicates, auditing missing values, and retaining missing values as NaN rather than imputing artificial values. This preserved the original 1,550 records and produced 1,445 complete-case records for the main analysis.

The feature engineering work created variables such as salary_iqr, salary_premium, employment_premium, stability_score, high-outcome labels, and the Career Value Index. CVI is an original decision-support contribution of this project because it combines overall employment, full-time permanent employment, median salary, and salary stability into one transparent program-level index.

The modelling work included Kruskal-Wallis testing, salary regression, classification, leakage-controlled classification, and K-Means clustering. Regression performance was evaluated using MAE, RMSE, and R-squared. Classification performance was evaluated using accuracy, precision, recall, F1-score, ROC AUC, and baseline comparison.

A key refinement was the leakage-controlled classification step. Preliminary classification results were treated cautiously because some engineered variables were closely related to the target labels. The final predictive evidence therefore used context-only variables such as year, university, school, and degree, with the high-salary model providing the strongest defensible result.

The CVI contribution was also validated statistically. Correlation tests showed that CVI was significantly related to relevant graduate outcome features, and sensitivity analysis showed that CVI remained stable under equal-weight, employment-heavy, salary-heavy, and stability-heavy alternatives.

Overall, the original work is not just the use of one model. It is a complete analytics pipeline that combines data cleaning, feature engineering, statistical testing, prediction, leakage control, clustering, CVI design, CVI validation, and stakeholder-focused interpretation.

Compared with a basic salary-only or descriptive-summary approach, this project adds hypothesis testing, leakage-controlled prediction, clustering, CVI construction, CVI validation, and CVI sensitivity analysis, allowing graduate outcomes to be interpreted as multidimensional program-level evidence rather than simple ranking.

Data Source and Dataset

The dataset used in this project is the Graduate Employment Survey - NTU, NUS, SIT, SMU, SUSS & SUTD dataset published by the Ministry of Education through data.gov.sg. The dataset is made available under the Singapore Open Data Licence version 1.0. The raw and processed datasets are included in the GitHub repository for reproducibility.

Table 3. Dataset overview

Item	Value
Number of records before cleaning	1550
Number of variables before feature engineering	12
Number of variables after feature engineering	23
Complete-case records for key outcomes	1445
Complete-case percentage	93.23
Time period	2013 to 2024
Universities	NTU, NUS, SIT, SMU, SUSS, SUTD
Number of schools	78
Number of degree program names	419
Unit of analysis	University-degree-year / program-year record

Inclusion and Exclusion Criteria

The analysis includes all available program-year GES records from 2013 through 2024 for NTU, NUS, SIT, SMU, SUSS, and SUTD. Records with non-numeric placeholders in outcome variables were retained in the raw dataset but excluded from model-ready complete-case analyses. No individual graduate-level data is used. Therefore, all results are interpreted at the program-year level.

Data Cleaning and Feature Engineering

Data cleaning preserved the original 1,550 program-year records. The main cleaning steps were standardizing column names, converting employment and salary columns from object type to numeric type, identifying non-numeric placeholders such as na and N.A., checking duplicates, and documenting missing values. The numeric conversion step identified 105 records with missing values across key employment and salary variables, equal to 6.77% of the dataset. These missing values were not imputed; complete-case records were used for modeling.

Table 4. Engineered feature summary

Feature	Definition	Purpose
salary_iqr	75th percentile salary minus 25th percentile salary	Measures salary spread.
salary_premium	Gross mean salary minus gross median salary	Indicates salary skew.

Feature	Definition	Purpose
employment_premium	Overall employment minus full-time permanent employment	Measures gap between general employment and stable employment.
stability_score	Inverse normalized salary spread	Represents salary stability.
career_value_index	Weighted composite of employment, stable employment, salary, and stability	Decision-support metric for multidimensional outcomes.

Career Value Index Design

Career Value Index (CVI) is introduced as a constructed program-level decision-support index. The purpose of CVI is to summarize multiple graduate outcome dimensions into a single interpretable score so that program outcomes are not evaluated using salary alone. CVI combines overall employment, full-time permanent employment, median salary, and salary stability.

$$\text{CVI} = 0.30(\text{Overall Employment Score}) + 0.30(\text{Full-Time Permanent Employment Score}) + 0.25(\text{Median Salary Score}) + 0.15(\text{Salary Stability Score})$$

The CVI components are normalized before weighting so that employment percentages, salary values, and salary stability can be combined on a comparable scale. The theoretical CVI range is 0 to 1. A higher CVI indicates stronger combined program-level employment, stable employment, salary, and salary-stability outcomes. A lower CVI indicates weaker combined outcomes. CVI is not an official ranking, causal score, or individual employability predictor.

The CVI weighting scheme is a transparent design choice rather than a fixed universal rule. Overall employment and full-time permanent employment receive the highest combined weight because graduate outcome quality should first reflect whether graduates are employed and whether they obtain stable jobs. Median salary is included as an important economic outcome, while salary stability is treated as a supporting component because it helps capture salary consistency. To avoid depending on one exact weighting choice, the study later tests equal-weight, employment-heavy, salary-heavy, and stability-heavy CVI alternatives through sensitivity analysis.

Research Hypotheses

Table 5. Research hypotheses by research question

RQ	Null Hypothesis (H0)	Alternative Hypothesis (Ha)
RQ1	H0: There is no meaningful year-level change in salary or employment outcomes from 2013 to 2024.	Ha: Salary or employment outcomes show meaningful year-level change from 2013 to 2024.
RQ2	H0: Employment, salary, and CVI outcomes do not differ significantly across universities.	Ha: At least one university differs significantly in employment, salary, or CVI outcomes.
RQ3	H0: Selected year, employment, salary-spread, and university features are not associated with median salary or CVI.	Ha: One or more selected features are associated with median salary or CVI.
RQ4	H0: Context-only features cannot classify high-outcome program-year records better than baseline.	Ha: Context-only features can classify at least one high-outcome target better than baseline.
RQ5	H0: Degree programs cannot be segmented into interpretable employability-	Ha: Degree programs can be segmented into interpretable

RQ	Null Hypothesis (H0)	Alternative Hypothesis (Ha)
	salary profiles.	employability-salary profiles.

Minimum Sample Size Computation and Adequacy by Research Question

The dataset contains 1,550 program-year records before cleaning and 1,445 complete-case records for the main analysis. Because the project uses the full available population of program-level GES records rather than drawing a small sample, the available sample size is considered sufficient for descriptive, statistical, regression, classification, and clustering analysis.

For a conservative minimum sample-size check, the standard 95% confidence-level sample size formula was used:

$$n = Z^2 p (1 - p) / e^2$$

where $Z = 1.96$ for a 95% confidence level, $p = 0.50$ for maximum variability, and $e = 0.05$ for a 5% margin of error.

$$n = (1.96^2 \times 0.50 \times 0.50) / 0.05^2; n = 384.16 \sim 385$$

Therefore, the minimum recommended sample size is approximately 385 records. The complete-case dataset of 1,445 records is well above this minimum requirement.

For regression analysis, Green's rule of thumb was also considered:

$$N \geq 50 + 8m$$

where m is the number of predictors. With approximately 8 candidate predictors, the minimum recommended sample size is:

$$N \geq 50 + 8(8) = 114$$

The complete-case dataset of 1,445 records is also well above this regression adequacy threshold.

Table 6. Sample size adequacy by research question

RQ	Analysis Type	Minimum Requirement Used	Available Records	Adequacy Decision
RQ1	Trend and descriptive EDA	385 records using 95% confidence sample-size check	1,445 complete cases	Adequate
RQ2	Group comparison / Kruskal-Wallis testing	385 records using 95% confidence sample-size check	1,445 complete cases	Adequate
RQ3	Regression and correlation	114 records using Green rule with approximately 8 predictors	1,445 complete cases	Adequate
RQ4	Classification modeling	385 records and 362 high-outcome records per target	1,445 complete cases	Adequate
RQ5	Clustering and segmentation	385 records using 95% confidence sample-size	1,445 complete	Adequate

RQ	Analysis Type	Minimum Requirement Used	Available Records	Adequacy Decision
		check	cases	

Statistical and Modeling Methods

Table 7. Methods and purpose

Method	Type	Purpose
EDA	Coverage, univariate, bivariate, and multivariate analysis	Understand distributions, missingness, trends, group differences, and correlation structure.
Kruskal-Wallis test	Non-parametric group comparison	Test university-level outcome differences at 5% significance.
Mann-Whitney pairwise tests with Holm adjustment	Post-hoc pairwise comparison	Report only meaningful university differences without overloading the final report.
Linear regression	Salary association model	Explain factors associated with median gross monthly salary using MAE, RMSE, and R-squared.
Logistic Regression, Decision Tree, Random Forest	Classification models	Classify high-employment, high-salary, and high-CVI records using leakage-controlled validation.
K-Means and PCA	Segmentation and visualization	Create interpretable employability-salary profiles and visualize cluster separation.
CVI sensitivity analysis	Robustness validation	Test whether CVI remains stable under alternative weighting schemes.

System Overview and Workflow

The system is a reproducible analytics pipeline that begins with the official Singapore data portal GES dataset and ends with decision-support outputs for students, universities, and policymakers. The pipeline integrates data ingestion, cleaning, feature engineering, EDA, statistical testing, modeling, clustering, CVI validation, and report-ready outputs.



Figure 1. End-to-end workflow of the proposed analytics system

Workflow Components

Table 8. Workflow components

Component	Description
Data Ingestion	CSV file from the official Graduate Employment Survey dataset.
Data Preprocessing	Numeric conversion, missing-value audit, duplicate check, complete-case preparation.
Exploratory Data Analysis	Coverage, univariate, bivariate, and multivariate analysis.
Feature Engineering	salary_iqr, salary_premium, employment_premium, stability_score, CVI, high-outcome labels.

Component	Description
Model Development	Regression, classification, clustering, and CVI robustness analysis.
Model Evaluation	MAE, RMSE, R-squared, F1-score, ROC AUC, silhouette score, correlation, top-quartile overlap.
Deployment Concept	GitHub-hosted reproducible notebook and report-ready outputs that can be refreshed when new GES data is released.

Tools and Technologies

The implementation used Python, Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Jupyter Notebook, and Git/GitHub. The analysis was designed as a reproducible notebook workflow using tabular data files and output folders for tables and figures.

Results

Model Performance Summary

Table 9. Model and validation performance summary

Analysis	Features Used	Validation Method	Metric 1	Metric 2	Key Observation
Salary regression	Year, salary_iqr, employment_premium, FT permanent employment, overall employment, university	75:25 train-test split	MAE = 333.86; RMSE = 476.90	R-squared = 0.58	Reasonable association model for salary outcomes.
High-salary classification	Context-only predictors: year, university, school, degree	Degree-grouped leakage-controlled split	F1 = 0.71	ROC AUC = 0.91	Strongest defensible predictive result.
K-Means clustering	Overall employment, FT permanent employment, median salary, salary_iqr	Standardized features; k = 2 to 8 tested	Silhouette = 0.32	PCA variance = 82%	Three interpretable employability-salary profiles.
CVI sensitivity	Alternative CVI weighting schemes	Correlation, rank overlap, cluster stability	Spearman = 0.98 to 1.00	Top-quartile overlap = 90.06% to 95.86%	CVI is robust under reasonable alternative weights.

Results by Research Question

RQ1: Employment and Salary Trends

The year-level trend analysis shows that program-level salary outcomes increased across the study period. Median gross monthly salary increased from 3,100 in 2013 to 4,400 in 2024. This result indicates a clear upward salary trend across program-year records, but it should be interpreted as a program-level descriptive pattern rather than an individual salary prediction.

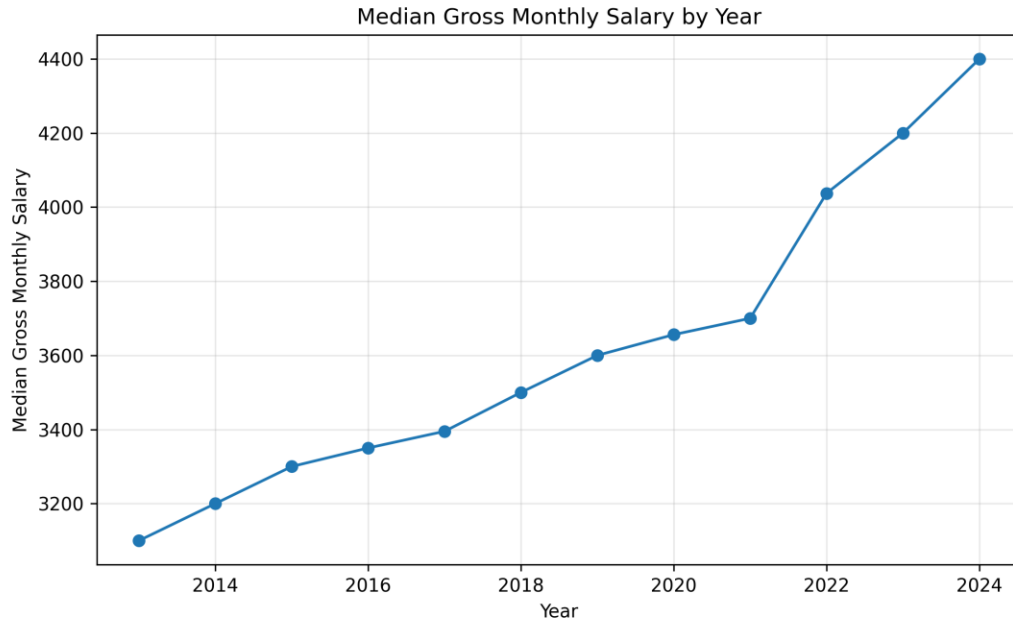


Figure 2. Median gross monthly salary trend by year

Employment outcomes remained generally high, but overall employment and full-time permanent employment did not move identically. Median overall employment reached 95.85% in 2021, while full-time permanent employment showed greater variation. The sharpest decline in full-time permanent employment occurred in 2020, when the median dropped to 70.50%, before recovering in 2021 and 2022.

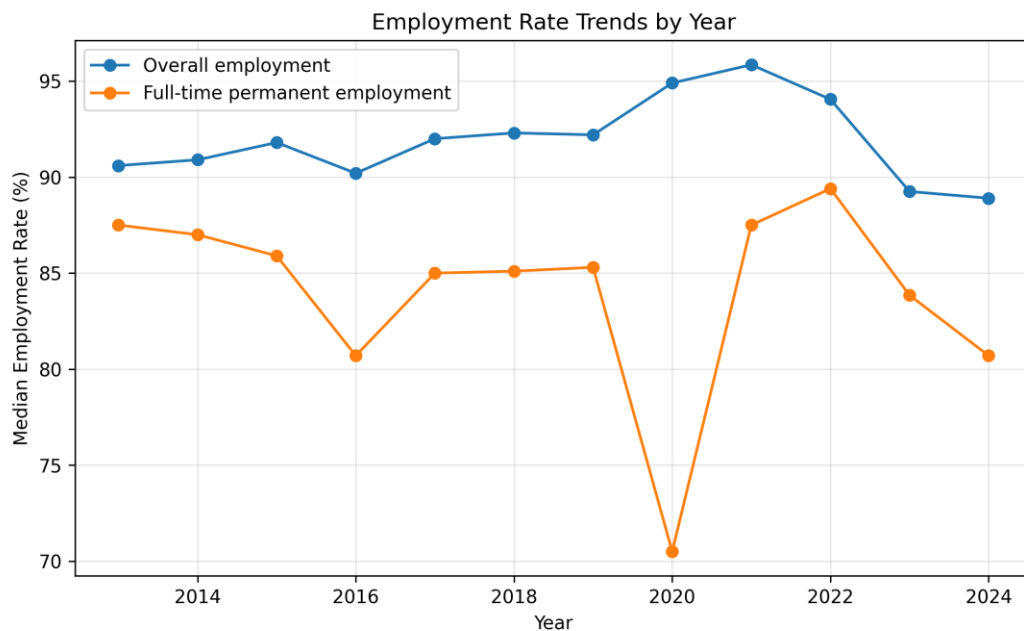


Figure 3. Overall and full-time permanent employment trends by year

RQ1 indicates that salary outcomes improved over time, but employment stability fluctuated more than overall employment. Therefore, salary growth should not be used alone to interpret graduate outcomes. Overall employment, full-time permanent employment, salary level, and salary stability should be considered together.

RQ2: University-Level Outcome Differences

Kruskal-Wallis tests found statistically significant university-level differences for all four tested outcomes: overall employment rate, full-time permanent employment rate, median gross monthly salary, and Career Value Index. The p-values were extremely small and are reported as $p < .001$. Effect sizes were modest, with epsilon-squared values from 0.047 to 0.067. This means university context matters, but it does not fully explain graduate outcomes.

Table 10. Statistical test summary

Outcome	Kruskal-Wallis H	p-value	Epsilon-squared	Interpretation
Overall employment rate	72	$p < .001$	0.047	Statistically significant university-level difference with modest effect size.
Full-time permanent employment rate	89.53	$p < .001$	0.059	Statistically significant university-level difference with modest effect size.
Median gross monthly salary	93.25	$p < .001$	0.061	Statistically significant university-level difference with modest effect size.
Career Value Index	101.85	$p < .001$	0.067	Statistically significant university-level difference with modest effect size.

Table 11. Selected meaningful pairwise university differences

Outcome	Higher Median	Lower Median	Median Difference	Adjusted p-value
Full-time permanent employment	National University of Singapore	Nanyang Technological University	6.7	$p < .001$
Full-time permanent employment	Singapore Institute of Technology	Nanyang Technological University	4.8	0.032
Full-time permanent employment	Singapore Management University	Nanyang Technological University	11.6	$p < .001$
Full-time permanent employment	Singapore University of Technology and Design	Nanyang Technological University	7.35	0.005
Median gross monthly salary	Singapore Management University	Nanyang Technological University	500	$p < .001$
Median gross monthly salary	Singapore University of Technology and Design	Nanyang Technological University	500	$p < .001$
Median gross monthly salary	Singapore Management University	National University of Singapore	300	$p < .001$
Median gross monthly salary	Singapore University of Technology and Design	National University of Singapore	300	0.019
Median gross monthly salary	Singapore Management University	Singapore Institute of Technology	500	$p < .001$
Career Value Index	National University of Singapore	Nanyang Technological University	0.045	$p < .001$

Outcome	Higher Median	Lower Median	Median Difference	Adjusted p-value
Career Value Index	Singapore Institute of Technology	Nanyang Technological University	0.043	0.011
Career Value Index	Singapore Management University	Nanyang Technological University	0.086	p < .001
Career Value Index	Singapore University of Technology and Design	Nanyang Technological University	0.062	p < .001
Career Value Index	Singapore Management University	National University of Singapore	0.041	p < .001

The selected pairwise results show that the most meaningful differences involve higher median salary patterns for SMU and SUTD relative to several other universities, higher full-time permanent employment for SMU and NUS relative to NTU, and higher CVI medians for SMU, NUS, SIT, and SUTD relative to NTU in selected comparisons. These results are reported selectively to avoid overemphasizing statistically significant but small differences.

RQ3: Salary and Career Value Drivers

The salary regression model used 1,445 complete-case records with a 75:25 train-test split. The model achieved MAE = 333.86, RMSE = 476.90, and R-squared = 0.58. Important associated factors included year, salary spread, employment premium, full-time permanent employment, overall employment, and university context. These results are associational and should not be interpreted causally.

Table 12. Regression performance summary

Model	Target	Validation	Records Used	MAE	RMSE	R-squared	Key Interpretation
Salary regression	gross_monthly_median	75:25 train-test split	1445	333.86	476.9	0.58	Reasonable program-level association model for median salary outcomes.

Table 13. Leading salary regression coefficient directions

Feature	Direction	Coefficient	Interpretation
num__year	Positive	391.588	More recent survey years are associated with higher predicted median salary.
num__salary_iqr	Positive	271.105	Programs with wider salary spread tend to have higher median salary, suggesting that higher-salary programs may also have wider salary distributions.
cat__university_Singapore University of Social Sciences	Negative	-251.469	Associated with lower predicted median salary relative to the model reference university, after other variables are included.
cat__university_Singapore University of Technology and Design	Positive	182.682	Associated with higher predicted median salary relative to the model reference university, after other variables are included.
cat__university_Singapore Institute of Technology	Negative	-147.629	Associated with lower predicted median salary relative to the model reference university, after other variables are included.

Feature	Direction	Coefficient	Interpretation
cat_university_Singapore Management University	Positive	136.451	Associated with higher predicted median salary relative to the model reference university, after other variables are included.
num_employment_premium	Negative	-134.676	A larger gap between overall employment and full-time permanent employment is associated with lower predicted median salary.
num_employment_rate_ft_perm	Positive	102.131	Higher full-time permanent employment is associated with higher predicted median salary.

The coefficient directions show that more recent years and salary-spread variables are strongly associated with higher median salary, while employment premium is negatively associated with median salary. The negative employment-premium association means that a wider gap between general employment and full-time permanent employment is linked to weaker stable employment context. University indicators capture remaining institutional context after numeric variables are included.

CVI correlation validation directly addresses whether the index can represent a meaningful set of features. Full-time permanent employment has a strong statistically significant positive relationship with CVI, overall employment also has a strong statistically significant positive relationship, and median gross monthly salary has a moderate statistically significant positive relationship. Employment premium is negatively related to CVI, which supports the stable-employment interpretation.

Table 14. CVI correlation validation evidence

Feature	Pearson r	Pearson p-value	Spearman rho	Spearman p-value	Strength
Full-time permanent employment	0.913	$p < .001$	0.935	$p < .001$	Strong
Overall employment	0.859	$p < .001$	0.869	$p < .001$	Strong
Median gross monthly salary	0.522	$p < .001$	0.505	$p < .001$	Moderate
Employment premium	-0.675	$p < .001$	-0.738	$p < .001$	Moderate
Salary stability score	-0.115	$p < .001$	-0.088	$p < .001$	Weak
Salary IQR	0.115	$p < .001$	0.088	$p < .001$	Weak

The weak negative relationship for salary stability is interpreted cautiously because the effect size is small and close to zero. This result should not be read as evidence that salary stability reduces career value. A more reasonable interpretation is that, in this dataset, some higher-salary programs also have wider salary dispersion. This is consistent with the salary regression results, where salary IQR was positively associated with median salary. Therefore, salary stability does not move as strongly with CVI as overall employment, full-time permanent employment, and median salary do. For this reason, salary stability is retained as a supporting component of CVI, but it is given a lower weight than the employment and salary-level components.

The Career Value Index diagnostic regression produced $R\text{-squared} = 1.00$ because the model included variables used to construct the index. This result is not independent predictive evidence. It is used only as a diagnostic consistency check. The stronger final contribution is the CVI sensitivity analysis, which tested whether the index remains stable under alternative weighting schemes.

Table 15. CVI sensitivity analysis results

CVI Version	Spearman rho	Spearman p-value	Top Quartile Overlap %	Mean Absolute Rank Change
Equal-weight CVI	0.995	$p < .001$	95.86	26.27
Employment-heavy CVI	0.997	$p < .001$	95.03	23.12
Salary-heavy CVI	0.987	$p < .001$	93.09	48.92
Stability-heavy CVI	0.984	$p < .001$	90.06	51.73

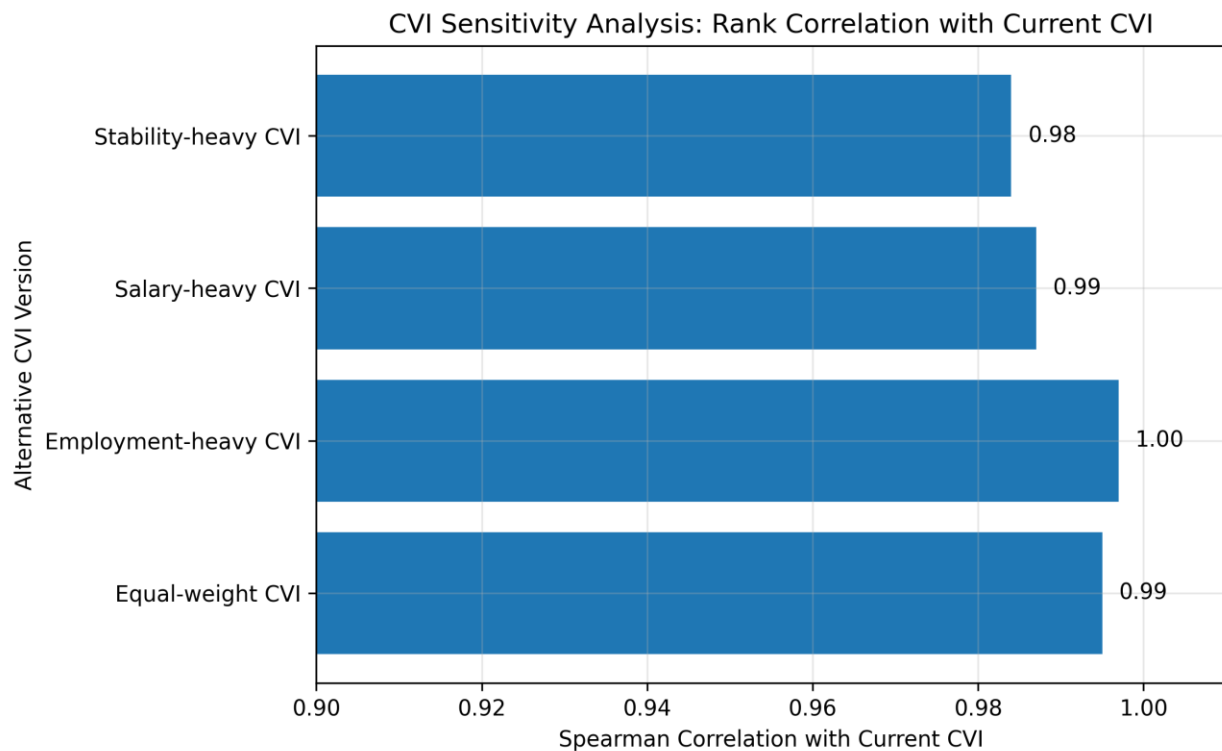


Figure 4. CVI sensitivity analysis: rank correlation with current CVI

Therefore, the CVI should be interpreted as a robust decision-support index rather than a weight-sensitive ranking, because the overall pattern remains stable even when reasonable alternative weights are applied.

The CVI sensitivity analysis shows that the proposed Career Value Index remains stable under reasonable alternative weighting schemes. Spearman rank correlations with the current CVI ranged from 0.98 to 1.00, and top-quartile overlap ranged from 90.06% to 95.86%. This supports CVI as a robust program-level decision-support index rather than a weight-sensitive ranking.

RQ4: Leakage-Controlled Classification Results

Preliminary classification models were developed for high employment, high salary, and high Career Value Index using top-quartile thresholds. The preliminary models performed strongly, but these results are likely optimistic because some engineered variables indirectly contained target information. To avoid overclaiming, the final report uses leakage-controlled classification as the main evidence. In the leakage-controlled design, only year, university, school, and degree were used as predictors.

Table 16. Preliminary and leakage-controlled classification performance

Target	Best Preliminary F1	Best Leakage-Controlled F1	Best Validation	Conservative	Key Takeaway
High employment	0.98	0.55-0.56	Logistic context-only	Regression,	Moderate performance after leakage control.
High salary	0.84	0.71-0.79	Logistic context-only	Regression,	Strongest conservative classification result; degree-grouped F1 = 0.71 and ROC AUC = 0.91.
High Career Value	0.99	0.61-0.69	Logistic context-only	Regression,	Moderate performance after leakage control.

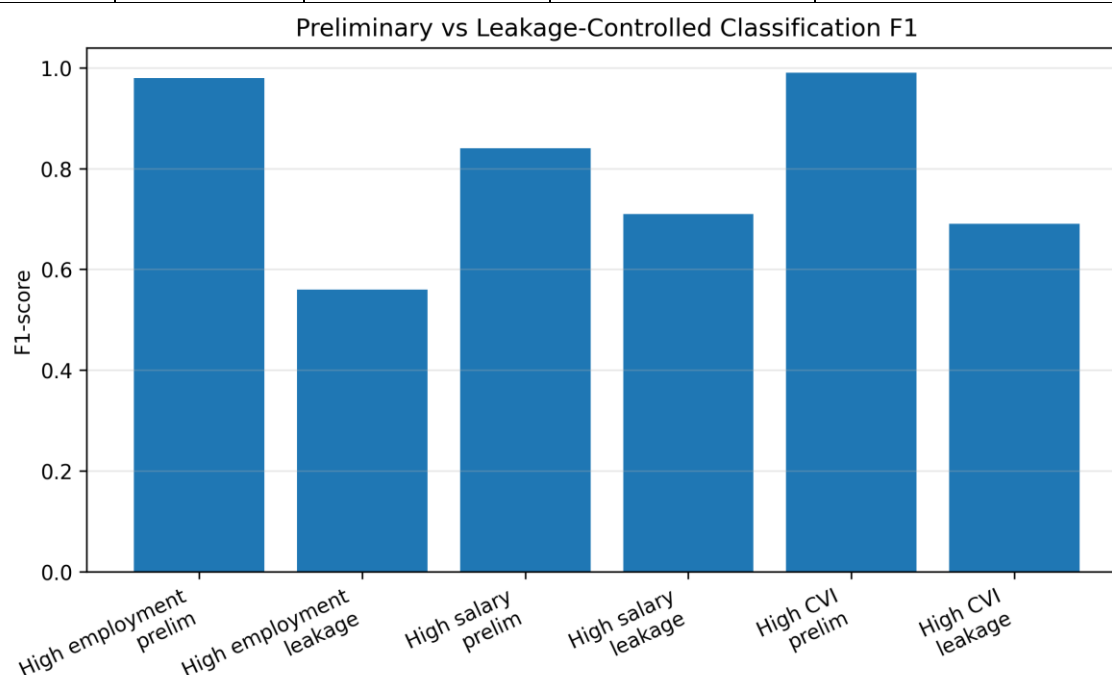


Figure 5. Preliminary versus leakage-controlled classification F1 scores

The leakage-controlled models produced more realistic results. High-salary classification was the strongest conservative task. Logistic Regression achieved $F1 = 0.79$ and $ROC\ AUC = 0.96$ under the random stratified context-only split. Under the stricter degree-grouped context-only split, the high-salary model achieved $F1 = 0.71$ and $ROC\ AUC = 0.91$. This is the most defensible predictive claim in the final report.

RQ5: Clustering and Segmentation Results

K-Means clustering was used to segment program-year records into employability-salary profiles. The clustering inputs were overall employment rate, full-time permanent employment rate, median gross monthly salary, and salary IQR. Career Value Index was not used as a clustering input; it was used only after clustering to interpret the resulting profiles.

To select the number of clusters, k values from 2 to 8 were evaluated using the K-Means elbow curve, silhouette score, PCA visualization, and practical interpretability. The three-cluster solution was selected because it provided a balanced segmentation without over-fragmenting the program-year records. While larger k values can reduce within-cluster error, $k = 3$ produced the clearest employability-salary profiles: high employment and high salary, high employment stability with moderate salary, and moderate or lower relative outcome. The silhouette score of 0.32 indicates moderate separation, which is reasonable for overlapping real-world graduate outcome data.

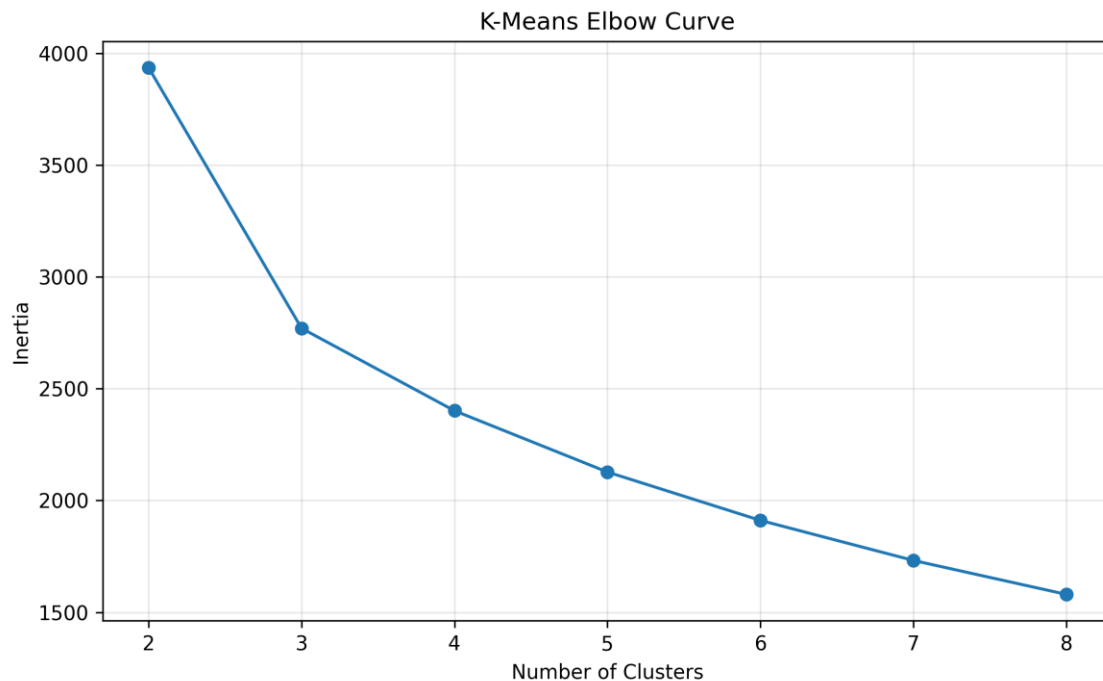


Figure 6. K-Means elbow curve

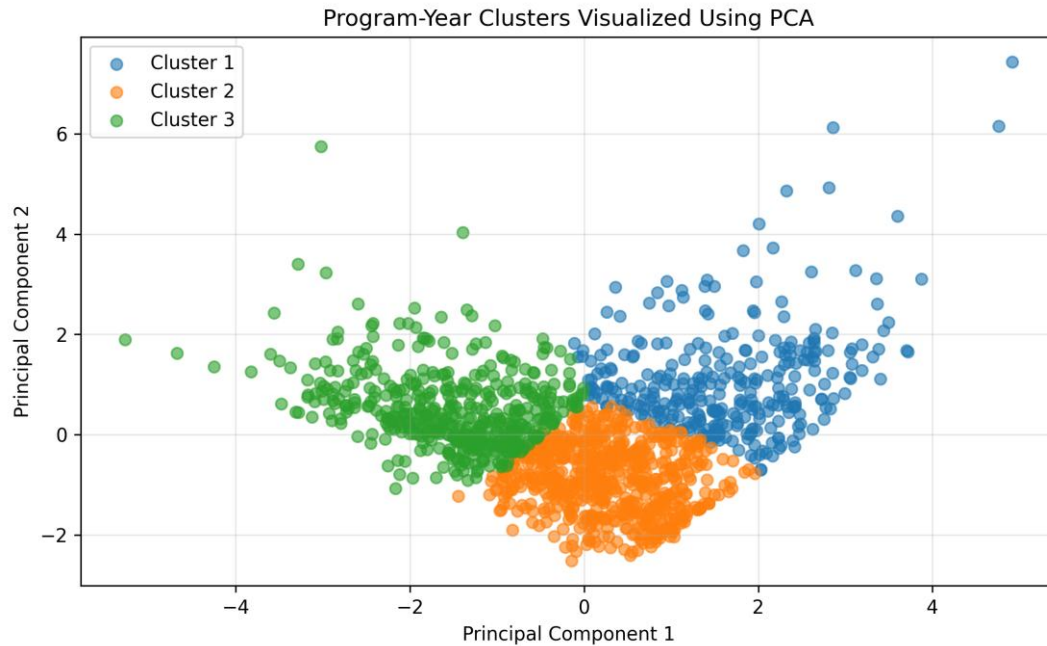


Figure 7. Program-year clusters visualized using PCA

Table 17. Cluster profile summary

Cluster	Record Count	Record %	Median Overall Employment	Median FT Permanent Employment	Median Salary	Median Salary IQR	Median CVI	Interpretation
1	324	22.42%	94.9	91.1	4,781.50	1,300.00	0.788	High employment and high salary profile
2	651	45.05%	94.9	89.2	3,500.00	600	0.743	High employment stability with moderate salary profile
3	470	32.53%	84.75	68.25	3,362.50	700	0.599	Moderate or lower relative outcome profile

The three clusters translate multiple quantitative indicators into interpretable program outcome groups. Cluster 1 represents high employment and high salary outcomes. Cluster 2 represents strong employment stability with moderate salary. Cluster 3 represents moderate or lower relative outcomes. The profiles support practical decision-making because they allow program comparison beyond salary alone.

Table 18. CVI cluster stability across weighting schemes

CVI Version	Cluster 1	Cluster 2	Cluster 3
Current CVI	0.788	0.743	0.599
Employment-heavy CVI	0.811	0.769	0.604
Equal-weight CVI	0.774	0.744	0.621
Salary-heavy CVI	0.751	0.687	0.565
Stability-heavy CVI	0.786	0.771	0.65

Across all CVI versions, Cluster 1 maintains the highest median CVI, Cluster 2 remains intermediate, and Cluster 3 remains the lowest relative profile. This reinforces the use of CVI as an interpretation tool after clustering rather than as a clustering input.

Overall Interpretation

The final results show that graduate outcomes in Singapore should be interpreted as multidimensional program-level outcomes rather than salary-only indicators. Salary outcomes increased from 2013 to 2024, but employment stability varied more across years. University-level differences were statistically significant but modest, which means institutional context matters but does not fully explain outcomes. Regression results show that salary is associated with year, salary spread, employment premium, full-time permanent employment, and university context. Classification results show that high-salary outcomes can be identified more reliably than high-employment or high-CVI outcomes when feature leakage is controlled. Clustering shows that programs can be grouped into interpretable employability-salary profiles. CVI adds decision-support value by summarizing multiple graduate outcome dimensions and remaining stable under alternative weighting schemes.

Practical Significance and Implementation

The results can be used in real-world decision-making by students, universities, and policymakers. Students can compare degree programs using salary, stable employment, salary spread, and CVI rather than salary alone. Universities can monitor programs that fall into lower relative outcome profiles and identify where career-services support or employer engagement may be useful. Policymakers can use the repeatable workflow to monitor program-level graduate outcomes when new GES data is released.

Implementation and User Benefit

Table 19. Implementation and user benefit

Target User	How the System Is Used	User Benefit
Students	Compare programs using salary, stable employment, salary spread, and CVI.	Better program selection beyond headline salary.
Universities	Monitor programs by cluster profile and CVI trend.	Identify programs needing career guidance, employer engagement, or communication support.
Policymakers	Refresh the workflow when new GES data is released.	Track labor-market alignment and outcome changes over time.
Analysts / evaluators	Use GitHub artifacts and notebook outputs.	Reproduce tables, figures, and model results.

Deployment Approach and System Integration

The project can be deployed as a reproducible Python/Jupyter notebook that generates report-ready tables and figures. The notebook can be rerun when a new GES dataset is published, producing updated tables, figures, CVI values, cluster profiles, and model summaries. These outputs can later be integrated into Tableau, Power BI, or a web-based dashboard for stakeholder review.

User Interaction and Example Use Case

A user would interact with the system by opening a dashboard or report generated from the latest processed data. For example, a student comparing two degree programs could review median salary, overall employment, full-time permanent employment, salary IQR, CVI, and cluster profile. A university administrator could filter programs by school or cluster and identify programs with high employment but moderate salary, or programs with lower full-time permanent employment. This allows the analysis to support practical interpretation rather than simple ranking.

Recommendations

Table 20. Final recommendations by stakeholder

Stakeholder	Recommendation	Application
Students	Use a multidimensional comparison view instead of salary-only comparison.	Review median salary, full-time permanent employment, salary spread, CVI, and cluster profile together.
Universities	Use cluster profiles to identify programs that may need additional career support.	Focus on programs with lower stable employment or moderate/lower relative outcome profiles.
Policymakers	Use the workflow as a repeatable monitoring framework.	Refresh analysis annually when new GES data is released and track shifts by program group.
Future dashboard users	Treat CVI as decision support, not ranking.	Use CVI to summarize outcome dimensions while checking underlying employment and salary variables.

Limitations and Further Improvements

Table 21. Limitations and mitigation plan

Area	Limitation/Risk	Impact	Mitigation
Data granularity	Dataset is program-year level, not individual-level.	Individual predictions cannot be claimed.	State all findings as program-level.
Causality	Analysis is observational.	Regression and classification are associational.	Avoid causal language and present results as data-driven evidence.
Missing values	105 records have missing key outcomes.	Models use 1,445 complete cases.	Report complete-case handling clearly.
Class imbalance	High-outcome class is about 25%.	Accuracy alone can mislead.	Report precision, recall, F1, ROC AUC, and baseline.
Feature leakage	Preliminary classification used related engineered variables.	Preliminary results are optimistic.	Use leakage-controlled results as main evidence.
Clustering interpretation	Clusters are descriptive profiles.	Profiles are not causal rankings.	Use clusters for decision support only.
CVI design weights	CVI weights are design choices.	Different users may value employment, salary, or stability differently.	Use sensitivity analysis and allow dashboard users to view alternative CVI versions.

Future Improvements

- Add time-based validation so models are trained on earlier years and tested on later years.
- Add more years of GES data as they become available.
- Develop an interactive dashboard where users can filter by university, school, degree, year, and cluster profile.
- Allow users to adjust CVI weights interactively to reflect different decision preferences.
- Incorporate external labor-market indicators if reliable program-level linkage becomes available.
- Improve classification validation using stronger temporal and program-family grouping designs.

Future Scope

The study can be extended into a reusable graduate-outcome monitoring tool. Future versions could support annual GES updates, dashboard-based decision support, interactive CVI weighting, automated report generation, and longitudinal monitoring of program-level outcome changes. The method could also be adapted to other countries or education systems if comparable program-level employment and salary data are available.

Conclusion

This final capstone project demonstrates that Singapore graduate employment and salary outcomes should be interpreted as multidimensional program-level outcomes. The analysis shows that median salary increased from 2013 to 2024, but stable employment varied more than

overall employment. University-level differences were statistically significant but modest. Salary regression provided a reasonable association model, while leakage-controlled classification showed that high-salary outcomes were the strongest defensible predictive target. K-Means clustering identified three practical employability-salary profiles. The Career Value Index was statistically validated and robustness-tested through sensitivity analysis, supporting its use as a transparent program-level decision-support index. Overall, the project provides a reproducible analytics framework that can support students, universities, and policymakers in interpreting graduate outcomes beyond salary alone.

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Appendix A: Output File and Figure/Table Path Guide

The appendix lists the main output files generated by the notebook. These files support the final report and provide the paths used for reproducibility. The appendix is referenced in the system overview, results, and practical implementation sections.

Table A1. Key figure paths

Figure	File Path	Report Section
Figure 1	outputs/figures/figure_01_end_to_end_workflow.png	System Overview and Workflow
Figure 2	outputs/figures/figure_07_01_salary_trend_by_year.png	RQ1 / Salary trend
Figure 3	outputs/figures/figure_07_02_employment_trend_by_year.png	RQ1 / Employment trend
Figure 4	outputs/figures/figure_cvi_sensitivity_spearman_correlation_zoomed.png	RQ3 / CVI robustness
Figure 5	outputs/figures/figure_13_02_preliminary_vs_leakage_controlled_f1.png	RQ4 / Classification comparison
Figure 6	outputs/figures/figure_14_01_kmeans_elbow_curve.png	RQ5 / K-Means cluster selection
Figure 7	outputs/figures/figure_14_03_cluster_pca_scatter.png	RQ5 / Cluster visualization
Figure B1	outputs/figures/appendix/figure_B1_combined_univariate_eda_summary.png	Appendix B / Univariate EDA
Figure B2	outputs/figures/appendix/figure_B2_combined_bivariate_eda_summary.png	Appendix B / Bivariate EDA
Figure B3	outputs/figures/appendix/figure_B3_multivariate_cvi_validation_summary.png	Appendix B / Multivariate and CVI validation

Table A2. Key table paths

Table Output	File Path	Use
Dataset overview	outputs/tables/dataset_overview.csv	Data Description
Data dictionary	outputs/tables/data_dictionary.csv	Data Dictionary / Appendix reference
Data cleaning log	outputs/tables/data_cleaning_log.csv	Data Cleaning
Engineered feature summary	outputs/tables/feature_engineering_summary.csv	Feature Engineering
Year trend summary	outputs/tables/bivariate_year_trend_summary.csv	RQ1
University outcome summary	outputs/tables/bivariate_university_outcome_summary.csv	RQ2
Statistical test results	outputs/tables/statistical_test_results.csv	RQ2
Pairwise university comparisons	outputs/tables/statistical_test_pairwise_comparisons.csv	RQ2
Regression performance	outputs/tables/regression_performance_summary.csv	RQ3
Regression coefficients	outputs/tables/regression_coefficient_summary.csv	RQ3
CVI correlation validation	outputs/tables/cvi_correlation_validation.csv	RQ3 / CVI validation
CVI weighting schemes	outputs/tables/cvi_weighting_scheme_table.csv	RQ3 / CVI design
CVI sensitivity analysis	outputs/tables/cvi_sensitivity_analysis.csv	RQ3 / CVI robustness
CVI cluster stability	outputs/tables/cvi_cluster_stability_analysis.csv	RQ5 / CVI interpretation
Preliminary vs leakage-controlled classification	outputs/tables/classification_preliminary_vs_leakage_controlled.csv	RQ4
Leakage-controlled classification performance	outputs/tables/leakage_controlled_classification_performance.csv	RQ4
Cluster profile summary	outputs/tables/cluster_profile_summary.csv	RQ5
Cluster profile simple	outputs/tables/cluster_profile_simple.csv	RQ5
Clustering k-selection summary	outputs/tables/clustering_k_selection_summary.csv	RQ5
PCA explained variance	outputs/tables/clustering_pca_explained_variance.csv	RQ5
Final research question results	outputs/tables/final_research_question_results.csv	Overall results summary
Final analysis evidence summary	outputs/tables/final_analysis_evidence_summary.csv	Final evidence tracking
Appendix B figure index	outputs/tables/appendix_b_figure_index.csv	Appendix figure tracking

Appendix B: Detailed Exploratory Data Analysis Outputs

Detailed EDA outputs are provided in Appendix B to support the main findings without overloading the main report body. The appendix combines univariate, bivariate, and multivariate evidence into summary figures that support dataset suitability, variable understanding, trend analysis, group comparison, feature selection, and CVI validation.

Figure B1. Combined Univariate EDA Summary

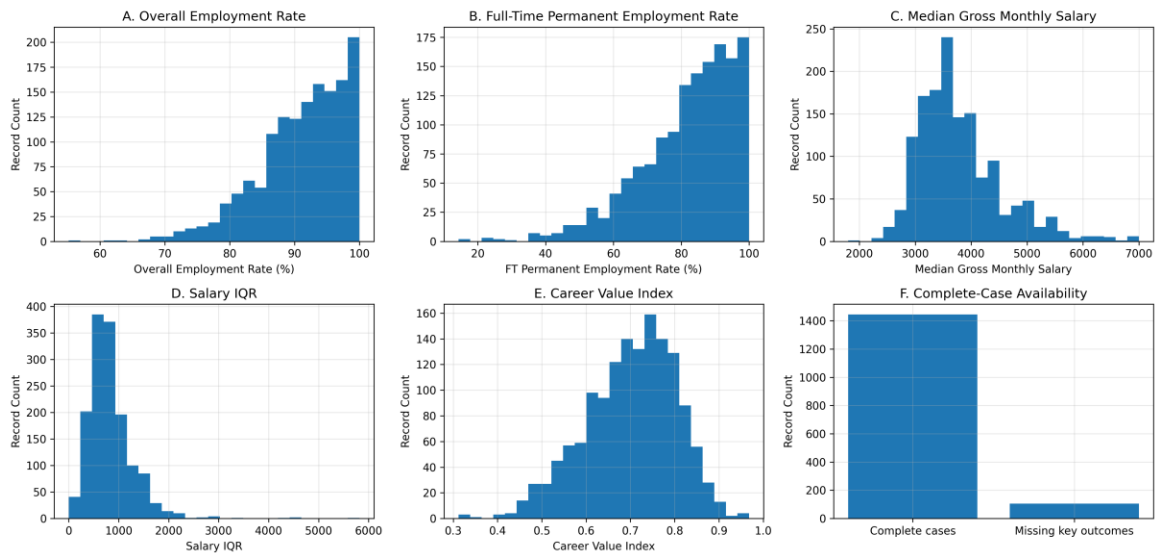


Figure B1. Combined univariate EDA summary of key employment, salary, salary-spread, and CVI variables

Figure B2. Combined Bivariate EDA Summary

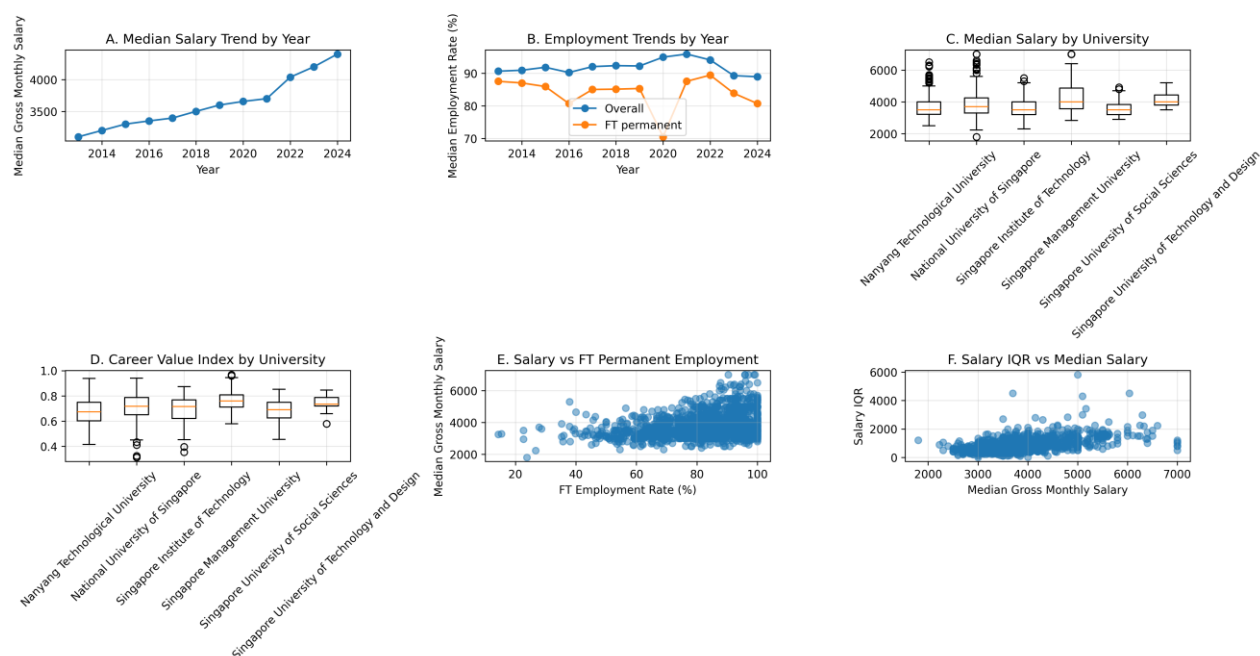


Figure B2. Combined bivariate EDA summary showing trends, university differences, and key relationships

Figure B3. Multivariate and CVI Validation Summary

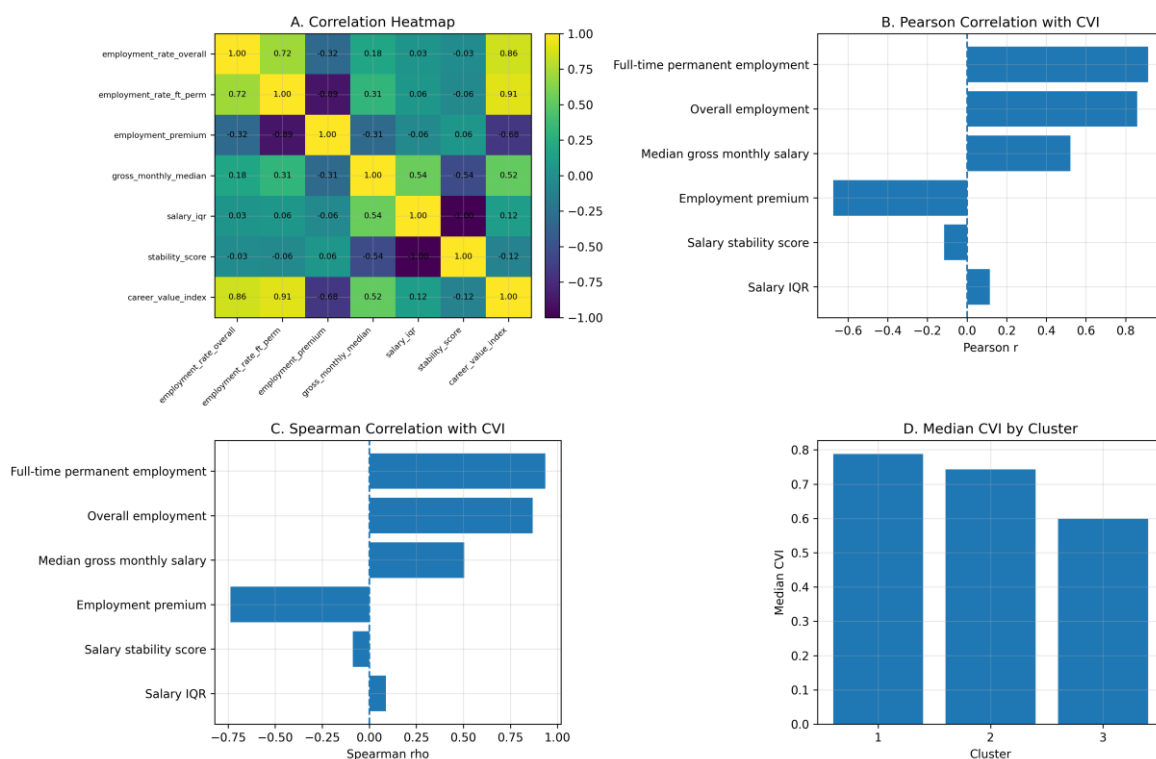


Figure B3. Multivariate and CVI validation summary

Appendix C: CVI Sensitivity and Robustness Outputs

Appendix C summarizes the CVI robustness checks used to test whether the proposed Career Value Index is overly dependent on one weighting scheme. The appendix includes the CVI weighting schemes, sensitivity analysis results, and cluster-level CVI stability results. These outputs support CVI as a transparent program-level decision-support index rather than an official ranking or causal score.

Table C1. CVI weighting schemes tested

CVI Version	Overall Employment Score	Full-Time Permanent Employment Score	Median Salary Score	Salary Stability Score
Current CVI	0.3	0.3	0.25	0.15
Equal-weight CVI	0.25	0.25	0.25	0.25
Employment-heavy CVI	0.35	0.35	0.2	0.1
Salary-heavy CVI	0.25	0.25	0.35	0.15
Stability-heavy CVI	0.25	0.25	0.2	0.3

Table C2. CVI sensitivity results for alternative weighting schemes

CVI Version	Spearman rho	Spearman p-value	Top Quartile Overlap %	Mean Absolute Rank Change
Equal-weight CVI	0.995	$p < .001$	95.86	26.27
Employment-heavy CVI	0.997	$p < .001$	95.03	23.12
Salary-heavy CVI	0.987	$p < .001$	93.09	48.92
Stability-heavy CVI	0.984	$p < .001$	90.06	51.73

Table C3. Cluster median CVI across weighting schemes

CVI Version	Cluster 1	Cluster 2	Cluster 3
Current CVI	0.788	0.743	0.599
Employment-heavy CVI	0.811	0.769	0.604
Equal-weight CVI	0.774	0.744	0.621
Salary-heavy CVI	0.751	0.687	0.565
Stability-heavy CVI	0.786	0.771	0.65